Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.07	20.27	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	2.03	+1.0%
power (W), mean	625	642	+2.7%
power (W), peak	648	682	+5.3%
energy (Wh)	209	217	+3.7%
energy, parked workers still on at idle power (Wh)	209	217	+3.7%
CPU used (cores), mean	0.780	1.23	+58.0%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.032	0.051	+58.0%
energy per core-hour (Wh)	802	521	-35.0%
node-hours per core-hour	7.70	4.87	-36.7%
pending pods, pod-minutes	0.500	0.000	-100.0%
pending pods, peak	1.00	0.000	-100.0%
HPA replicas, mean	8.59	9.41	+9.5%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.517	-34.0%
requests timed	1,130	1,100	-2.7%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	65.51	95.69	+46.1%
response time (ms), median	45.10	81.35	+80.4%
response time (ms), 95th percentile	156	156	-0.3%
response time (ms), 99th percentile	243	177	-27.0%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	8.6082	5.5950	-35.0%	[-5.1760, -0.9632]	GAIN
kwh_per_core_hour	0.8926	0.5929	-33.6%	[-0.5070, -0.0990]	GAIN
utilisation	0.0322	0.0505	+57.0%	[+0.0058, +0.0321]	GAIN
pending_pod_minutes_per_hour	1.4575	0.0000	-100.0%	[-3.6408, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.3352	1.5238	-34.7%	[-4.6952, +2.2857]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.359 0.392 0.421 0.446 0.468 0.488 0.504 0.52 0.534 0.546 0.556 0.561 0.566 0.571 0.576 0.581 0.585 0.588 0.589 0.591
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	613	619	4.00	6.00	0.000	0.000	0.450	0.632

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	613	624	4.00	6.50	0.000	0.000	0.429	0.774
2	6.00	6.00	615	624	4.00	8.00	0.000	0.000	0.467	0.756
3	6.00	6.00	616	631	4.00	8.00	0.250	0.000	0.557	0.953
4	6.00	6.00	629	644	7.50	10.00	0.000	0.000	0.875	1.32
5	6.00	6.00	628	650	9.00	10.00	0.000	0.000	0.876	1.45
6	6.00	6.00	630	652	10.00	10.00	0.250	0.000	0.918	1.50
7	6.00	6.00	640	672	10.00	10.00	0.000	0.000	1.19	2.04
8	6.00	6.00	642	677	10.00	10.00	0.000	0.000	1.23	2.15
9	6.00	6.00	643	673	10.00	10.00	0.000	0.000	1.27	2.07
10	6.00	6.00	641	666	10.00	10.00	0.000	0.000	1.18	1.86
11	6.00	6.00	616	623	10.00	10.00	0.000	0.000	0.545	0.727
12	6.00	6.00	616	625	10.00	10.00	0.000	0.000	0.509	0.776
13	6.00	6.00	622	628	10.00	10.00	0.000	0.000	0.685	0.886
14	6.00	6.00	632	653	10.00	10.00	0.000	0.000	0.985	1.52
15	6.00	6.00	628	654	10.00	10.00	0.000	0.000	0.859	1.55
16	6.00	6.00	632	645	10.00	10.00	0.000	0.000	0.952	1.32
17	6.00	6.00	622	639	10.00	10.00	0.000	0.000	0.669	1.17
18	6.00	6.00	614	622	10.00	10.00	0.000	0.000	0.498	0.720
19	6.00	6.00	615	624	10.00	10.00	0.000	0.000	0.522	0.772
20		6.00		627		10.00		0.000		0.806

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.